

- 4 A pike is a fish found in freshwater rivers.

The photograph shows a male and a female pike.



(Source: © Ross Jolliffe/Alamy Stock Photo)

Female pike lay egg cells on the leaves of river plants.

Male pike then release sperm cells that will fertilise the egg cells.

- (a) (i) An unfertilised egg cell is 2.3 mm in diameter.

Calculate the volume of the egg cell, using the formula:

$$V = \frac{4}{3}\pi r^3$$

Give your answer to **one** decimal place.

(1)

Answer mm³

- (ii) An unfertilised pike egg cell has a larger volume than a pike sperm cell.

Explain why an egg cell has a larger volume than a sperm cell.

(2)

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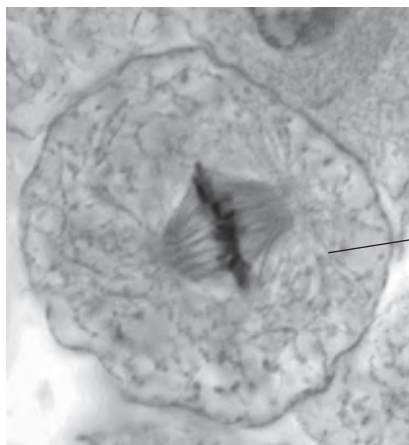
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(b) The photograph shows a cell from the blastocyst stage of development.

The blastocyst cell was treated to show the stage of mitosis.



cell from blastocyst

(Source: © HERVE CONGE, ISM/SCIENCE PHOTO LIBRARY)

(i) Which stage of mitosis is shown in this blastocyst cell?

(1)

- ☐ **A** cytokinesis
- ☐ **B** metaphase
- ☐ **C** prophase
- ☐ **D** telophase

(ii) State how the blastocyst cell was treated to show the stage of mitosis.

(1)



(iii) A blastocyst develops from a morula.

Compare and contrast the structure of a morula with the structure of a blastocyst.

(3)

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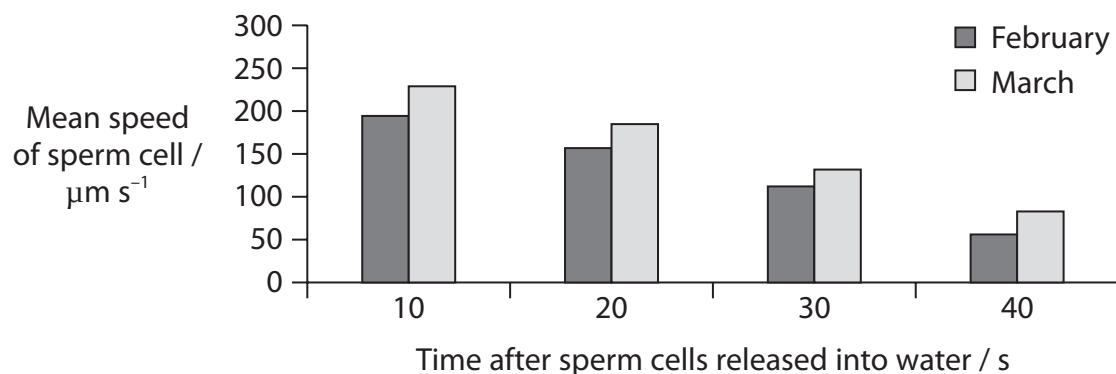
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(c) Pike sperm cells begin swimming when they are released into water.

A scientist investigated the speed of the sperm cells after they were released into water during February and March.

The graph shows the results of this investigation.



(i) Describe the adaptations of a sperm cell that allow movement.

(2)

(ii) Give **two** conclusions for this investigation.

(2)

1

2

(Total for Question 4 = 12 marks)

